ORF307: Optimization

Precept: #5

Agenda

- Converting LPs
- Steel company operations
- The moment problem (Optimization helps Probability Theory)
- Chebyshev center (maximum inscribed ball)

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW: ....
- Other announcements on blackboard.

Converting LPs

Let $A \in \mathbb{R}^{m\times n}, x \in \mathbb{R}^n, b \in \mathbb{R}^m$, and $k \in \mathbb{R}$.

We would like to convert the constrined optimization problem

$$\begin{aligned} \min_x \quad & \|Ax - b\|_1 \\ s.t. \quad & \|x\|_\infty \leq k \end{aligned} \tag{1}$$

into the following two formulations:

- Formulation 1:

$$\begin{aligned} \min_{\tilde{x}} \quad & \tilde{c}^T\tilde{x} \\ s.t. \quad & \tilde{A}\tilde{x} = \tilde{b} \\ & \tilde{C}\tilde{x} = \tilde{d} \end{aligned}$$

- Formulation 2:

$$\begin{aligned} \min_x \quad & c^Tx \\ s.t. \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

Solution:

1 Steel company operations

- A steel company can produce bands and coils
- The goal is to maximize its revenue
- The available data is shown in the following table

Table 1: Data for the steel company.

	Production rate (tons/hr)	Revenue ($/ton)	upper bounds (tons)
Bands	200	25	6000
Coils	140	30	4000

- There are 40 hours of production time this week
- Decide how many tons of bands and coils should be produced to maximize revenue.

Solution:

Moment problem

This problem shows how to use linear programming to find the maximum and minimum possible values of a random variable's moment, given some known moments. This is useful for bounding quantities when we have incomplete information about a probability distribution. The definition of the moments of a random variable is discussed in "ORF309-Probability and Stochastic Systems".

- Suppose that Z is a discrete random variables taking values in the set $\{0, 1, \dots, k\}$ with probabilities $p_0, p_1, \dots, p_k$
- We know the values of the 1^{st} and 2^{nd}, that is $\mathbb{E}[Z] = m_1$ and $\mathbb{E}[Z^2] = m_2$.

  We note that these moments can be defined in terms of the probabilities p_i as follows:

$$\mathbb{E}[Z] = \sum_{i=0}^{k} p_i i \quad \text{and} \quad \mathbb{E}[Z^2] = \sum_{i=0}^{k} p_i i^2$$

- We would like to obtain upper and lower bounds on the 4^{th} moment defined as

$$\mathbb{E}[Z^4] = \sum_{i=0}^{k} p_i i^4$$

- Show how LPs can be used to approach this problem.

Solution:

Chebyshev center

- Consider a set P described by linear inequality constraints

$$P = \{x \in \mathbb{R}^n | a_i^T x \leq b_i, i = 1, 2, \ldots, m\}$$

  Such a set is called a polyhedron.

- **Goal 1**: find a ball with the largest possible radius that is entirely contained within the polyhedron P.

- **Goal 2**: Provide an LP formulation of this problem.

Solution: